

Chapter 1

Introduction to Econometrics and Linear Regression

This chapter introduces a foundational concept in Econometrics. The primary focus of this chapter is to develop theoretical concepts including the different types of Econometric data, essential modeling techniques, underlying mathematical principles, and the critical role of hypothesis testing in Econometric models.

1. Econometrics

Econometrics integrates economic theory, mathematical modeling, and statistical inference to analyze real-world economic data. It provides a structured approach to understanding complex economic phenomena, enable the exploration of topics such as consumer behavior, economic growth, inflation, market dynamics, etc. It is a subject of quantitative analysis and by quantifying relationships between variables, Econometrics helps in making predictions and policy decisions. In Econometrics studies, researchers develop models based on theoretical relationships, define hypotheses and subsequently estimate, and test those models using statistical techniques. By doing so, they aim to validate economic theories or develop new data driven insights that can guide decision-making in policy, finance, and business.

Key methods in Econometrics include regression analysis for exploring relationships between variables. Regression in Econometrics is generally performed by taking different data types and approaches. These involve time dependent data for examining the trends over time, panel data analysis for handling datasets with both cross-sectional and time dimensions, and causal inference techniques for identifying cause-and-effect relationships.

The history of Econometrics started with Frisch (1926)¹, who came up with the term “Econometric” in his first economic publication. Empirical studies show that, while Tinbergen² popularized the Econometrics theory, Frisch was primarily responsible for its creation. Tinbergen (1936 & 1937) made a significant contribution to Econometrics by developing the first aggregate-level Econometric model. Later, [Klein, 1969](https://en.wikipedia.org/wiki/Johannes_Klein)^[1] and [Simon Kuznets, 1971](https://en.wikipedia.org/wiki/Simon_Kuznets)^[2] also contributed significantly to the Econometric research. All of them won the Nobel Prize in

¹ https://en.wikipedia.org/wiki/Ragnar_Frisch

² https://en.wikipedia.org/wiki/Jan_Tinbergen

economics for their contributions. While Klein is popular for his work in creating models to forecast economic trends, Simon Kuznets pioneered the notion of Gross Domestic Product (GDP), which captures all economic activity in a state using a single metric.

In Econometric approach, the first step is to obtain and analyze a dataset and define hypotheses based on the existing economic theories. Let us take a simple example to better understand this:

- Assuming a hypothetical scenario where we wish to find the correlation between the monthly price change of the “Tesla” stock price and the “Unemployment Rate” in the USA. So, we would collect both the data sets and define a hypothesis that “*Hypothesis* → *higher unemployment leads to lower Tesla stock prices.*” Here, price is the dependent or response variable, and unemployment rate is the independent or explanatory variable.

Let us model our problem. If we can model this successfully, we can run the regression and test the hypothesis to establish the findings. A model is a simplified mathematical representation of the relationships between different variables. Modeling helps clarify the structure of the relationship.

While assessing the relationship between two variables, it is critical to understand how they are related. The relationship here can be explored with a model defining a simple equation based on linear algebra, such as $Y = \beta_0 + \beta_1 X + \epsilon$ where, Y is dependent variable, X is the independent variable, β_0 is the slope or gradient of the line, β_1 is the intercept, and ϵ is the error. This represents a classic Econometric model.

In statistics, a linear relationship is a straight line between two variables shown in the above equation. Usually, when we look at the trend in a time dependent dataset (time-series), we get the impression that there is a true linear relationship. A trend refers to the direction or pattern in which a variable moves over time, whether it is increasing, decreasing, or staying constant. For instance:

- If a stock price is rising over time, this indicates an upward trend. In this case, higher highs and higher lows are typically observed.
- If the stock price is consistently falling over time, it shows a downward trend, characterized by lower highs and lower lows.

- When there is no significant upward or downward movement, the price might fluctuate within a certain range, indicating a sideways or horizontal trend.

Detecting trends is essential because they can influence forecasting and decision-making processes. Now going back to the model; a simple regression model generates a best-fit line through the data points in a scatter plot and calculates the average distance of each data point from this line. This is, known as error estimation. This line of best fit, or trendline, represents the estimated position of a linear equation within the dataset.

Several explanatory variables can be added to the analysis, such as changes to “GDP” and “inflation” in addition to “unemployment”, to explain the stock prices. In the case where we use more than one explanatory variable, it is known as multiple, multivariate, or multifactor linear regression. Multifactor regression is the most used approach in Econometrics. The common approach to employing multifactor regression is the Ordinary Least Squares (OLS) regression, which can be performed on several types of cross-sectional or time-series data.

In the following sections, we will understand different types of data that can be used for Econometric modeling. The GitHub link of the Python notebook for this chapter is available at the **footnote**³.

1.1 Importance of modeling in Econometrics

We already discussed the significance of Econometric modeling. To comprehend the importance of models, let us examine a few fundamental ideas. Models provide a structured method for testing hypotheses, quantifying relations, forecasting future trends, and making evidence-based decisions in economic policy and business strategy.

The primary reasons for learning and using Econometric models are:

- Models evaluate economic ideas and challenge assumptions. Without empirical testing, economic concepts remain hypothetical.
- Models allow for the quantification of relationships between variables. For example, models measure how changes in interest rates affect inflation or how government

³ https://github.com/saritmaitra/PracticalGuideEconometrics/blob/main/Chapter_1.ipynb

spending impacts GDP. This quantification is essential for making informed policy decisions or business strategies.

- A major application of models is forecasting. Businesses use them to predict future sales, demand, or market trends, while governments use them for predicting economic growth or unemployment rates. These forecasts are critical for strategic planning and policy decisions.
- Models are essential to evaluate the effectiveness of the policies. For instance, a government might use an Econometric model to assess the impact of a tax policy on income inequality or employment. This helps in understanding whether the policy is meeting its intended goals.

1.2 Types of data

Various types of data are normally used in an Econometrics model.

- 1) Time series data where data collected over time and provide information about the variables in the dataset. **Table 1** shows a sample Python code snippet to generate synthetic data and a subsequent line plot with daily frequency.

Table 1. Time series line plot.

```
dates = pd.date_range(start = '2023-01-01', periods = 100, freq = 'D')
values = pd.Series(range(100))
data = pd.DataFrame({'Date': dates, 'Value': values})
data.set_index('Date', inplace = True)
plt.figure(figsize = (10, 6))
plt.plot(data.index, data['Value'], marker = 'o', linestyle = '-')
plt.title('Time Series Data')
plt.xlabel('Date'); plt.ylabel('Value')
plt.grid(True)
plt.show()
```

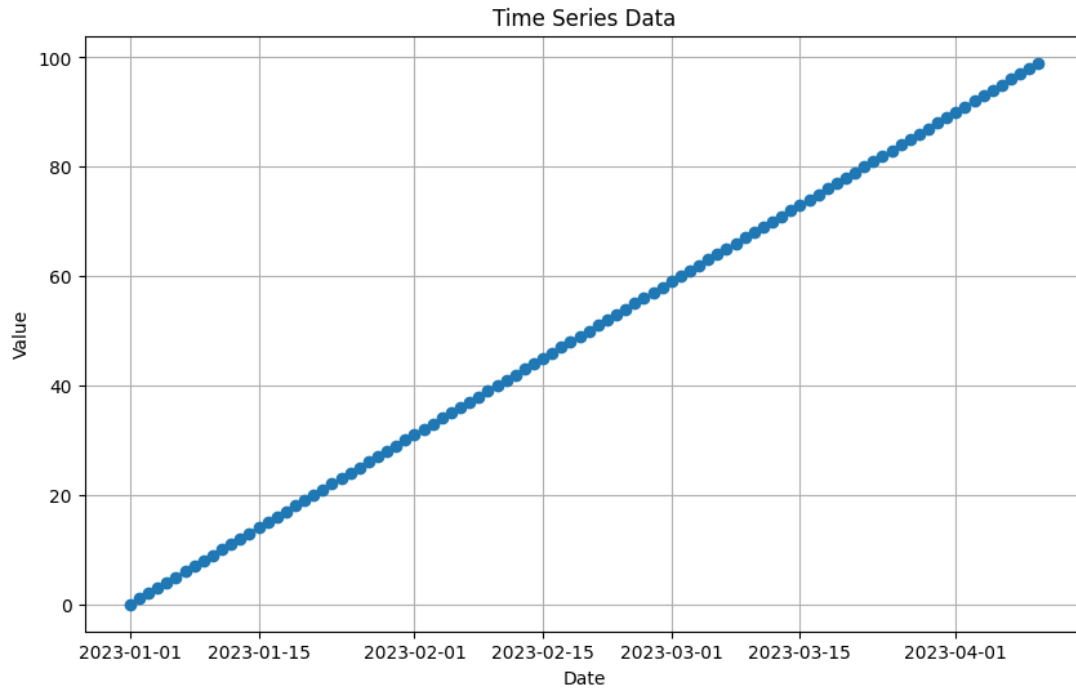


Figure 1. Line plot of Time series data.

2) Cross-sectional data refers to data collected by observing many subjects (such as individuals, companies, or countries) at one specific point in time, without considering how things change over time. **Table 2** shows a simple Python code to generate cross sectional data and a subsequent scatter plot.

Table 2. Synthetic cross-sectional data.

```
data = {
    'Individual': ['A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J'],
    'Value': [5, 7, 8, 5, 6, 7, 8, 7, 6, 5]
}
df = pd.DataFrame(data)
plt.figure(figsize = (10, 6))
plt.scatter(df['Individual'], df['Value'], marker = 'o', color = 'b')
plt.title('Cross-Sectional Data')
plt.xlabel('Individual');
plt.ylabel('Value')
plt.grid(True)
plt.show()
```

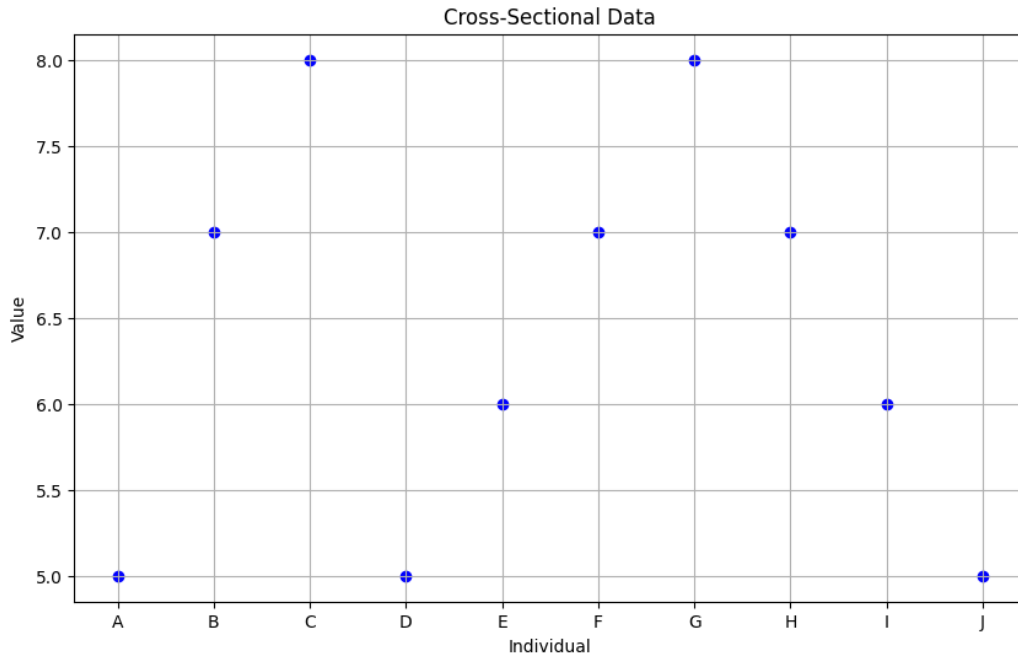


Figure 2. Sample scatter plot on cross sectional data.

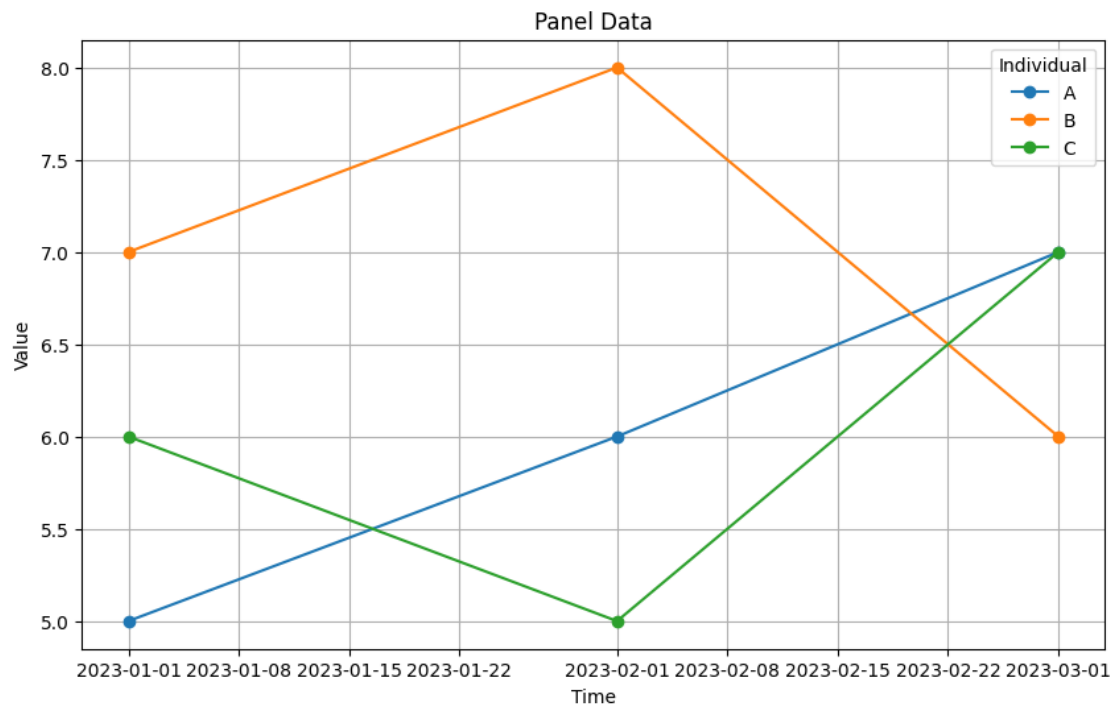
3) Panel data is a subset of longitudinal data in which the same subjects are observed repeatedly. Time series and cross-sectional data are special examples of panel data with a single dimension. **Table 3** shows simple Python code to generate synthetic panel data, and a subsequent plot displays the visualization.

Table 3. Synthetic panel data.

```
data = {
    'Individual': ['A', 'A', 'A', 'B', 'B', 'B', 'C', 'C', 'C'],
    'Time': ['2023-01', '2023-02', '2023-03', '2023-01', '2023-02', '2023-03', '2023-01', '2023-02', '2023-03'],
    'Value': [5, 6, 7, 7, 8, 6, 5, 6, 7]
}
df = pd.DataFrame(data)
df['Time'] = pd.to_datetime(df['Time'])

plt.figure(figsize = (10, 6))
for individual in df['Individual'].unique():
    subset = df[df['Individual'] == individual]
    plt.plot(subset['Time'], subset['Value'], marker = 'o', label = individual)
plt.title('Panel Data')
plt.xlabel('Time'); plt.ylabel('Value')
plt.legend(title = 'Individual')
plt.grid(True)
```

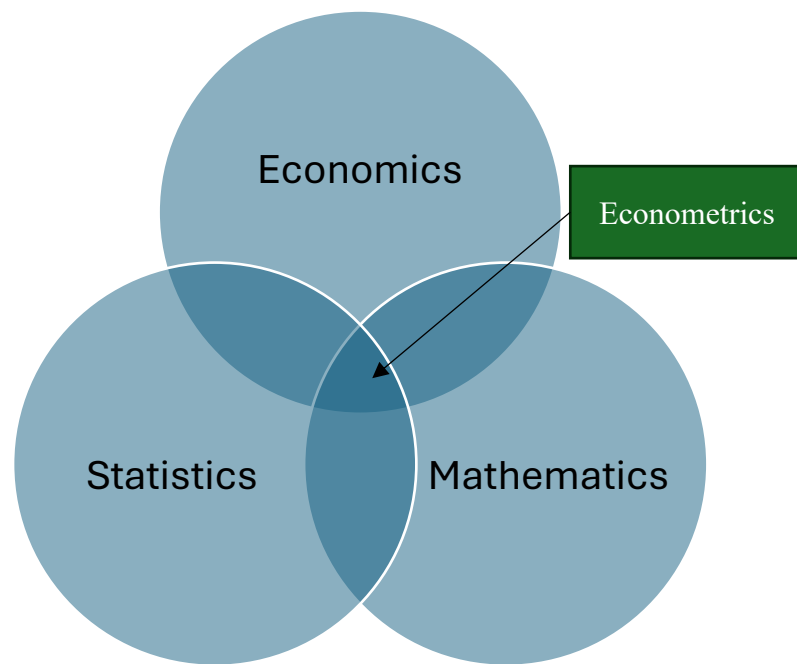
```
plt.show()
```



- 4) When the variables are qualitative in nature, the data are recorded in the form of the indicator function. These are dummy variables. Dummy variables are typically binary in nature which means they reflect only the presence or absence of a characteristic. For example, these values can be represented as ‘1’ being male and ‘0’ being female.

*A fundamental question may arise: “**what is the difference between economics and Econometrics?**”*

Econometrics is the intersection of three distinct fields: economics, statistics, and mathematics, as depicted in the Venn diagram below.



We already have discussed that Econometrics model is a quantitative examination of economic phenomena that employs mathematical models to evaluate economic theories and hypotheses. In this context, it is to be noted that, in quantitative finance, Econometric models have traditionally been applied to smaller datasets, and they are often used when researchers come up with a strong theoretical basis for the relationships being studied.

1.3 Key components

Figure 3 displays the key components of Econometric model development using a simple workflow.

Let us understand each of the steps shown in the workflow:

- I. Data collection involves gathering relevant economic data, such as historical prices, consumer survey results, unemployment rates, GDP data, inflation data, etc.
- II. A hypothesis proposes a solution for a specific problem or question that researchers must test. In other words, it is a testament that researchers test through research or experimentation to decide its validity.
- III. Model specification involves choosing a model that is the hypothesized relationships.
- IV. Estimation involves using statistical techniques to estimate the parameters of the chosen model. This often involves finding the best-fit line or curve that is the relationship between variables.

- V. Inference is combining parameter estimation and hypothesis testing to make inferences about the relationships between the variables. This includes assessing the statistical significance and strength of the estimated relationships.

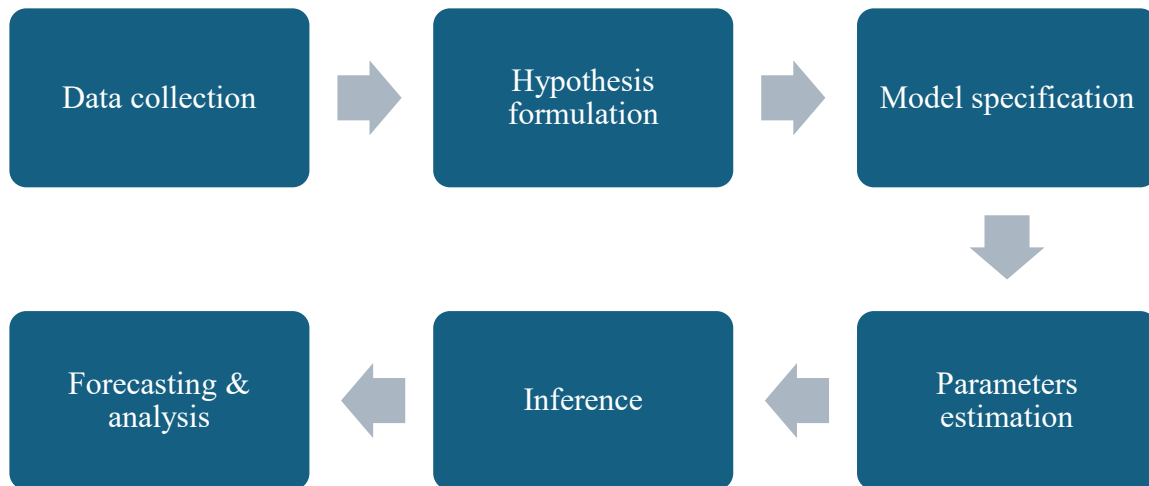


Figure 3. Econometric workflow

The final stage involves using the tested model to estimate future economic patterns and assess the possible impact of various policy interventions.

Various statistical techniques are involved in the modeling process to understand the data and subsequently process the data for robust modelling. Data cleaning and processing is a major activity for a model and reliable estimates. At a broader level, this often involves time frequency mapping, missing data handling, noise reduction, variable identification, frequency distributions, probability distributions, correlation analysis, and regression analysis.

- Frequency distributions are a summary of how often different values occur within a dataset. It is generally displayed as a table or histogram showing the frequency of each value or range of values.
- Probability distributions provide the probability of the occurrence of different outcomes. The normal distribution (Gaussian distribution) is the common probability distribution which displays that data near the mean are more frequent in occurrence than data far from the mean.

- Correlation analysis measures the strength and direction of the relationship between two variables. The correlation coefficient ranges from -1 to 1, where 1 means a perfect positive correlation, -1 means a perfect negative correlation, and 0 means no correlation.
- Regression analysis is commonly used to estimate variable relationships. It is a statistical technique to examine the relationship between one dependent variable and one or more independent variables.

Figure 4 expands the Econometric workflow by outlining a conceptual framework.

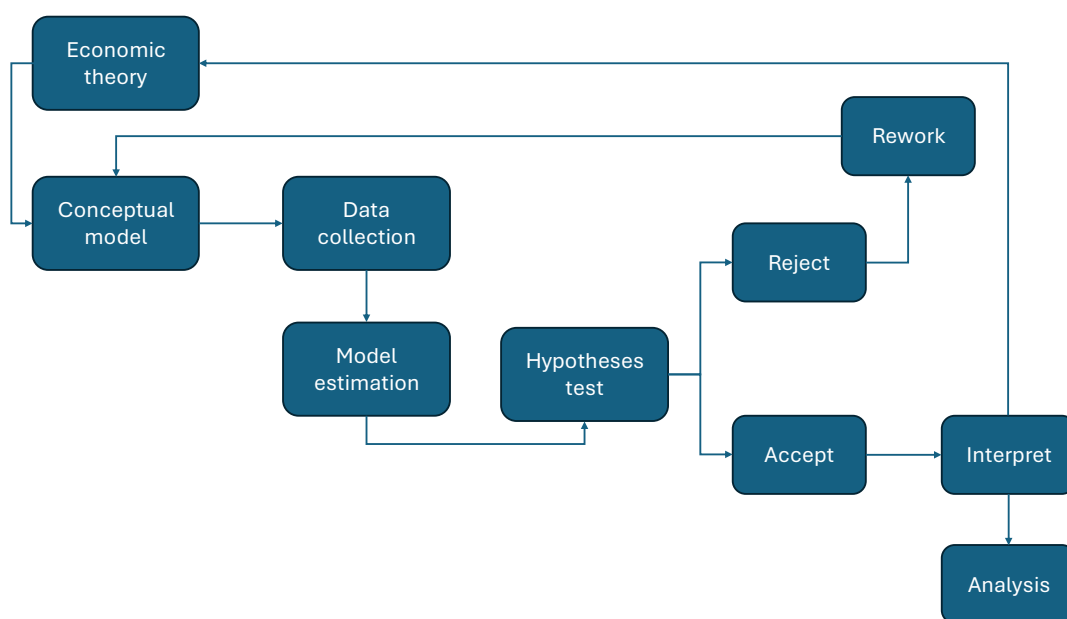


Figure 4. Different steps in Econometric modeling.

The process starts with formulating a theory of the problem. We can obtain the required data from published government sources. For estimation, either a univariate or a multivariate modeling equation can be used. The model's optimal parameter estimation relies on satisfying statistical assumptions and describing the data. Multiple hypotheses and validation processes are involved at this stage. If the assumptions are violated, we need to reformulate the work, collect more data, or choose a less stringent estimation technique. The hypothesis testing is done from a theoretical perspective. Once we are satisfied with the model, then use it to test the theory. Developing a robust model is an iterative process, and the final model may differ significantly from the original proposal.

1.4 Limitations

In applied economics, the biggest challenge is the lack of data for testing theories or hypotheses. Measurement errors and data revisions are also common issues. Measurement error occurs when the variables we use in our analysis do not perfectly measure the theoretical construct it is supposed to. This can lead to biased and inconsistent parameter estimates, which can compromise the validity of Econometric analyses. Measurement errors can affect both dependent and independent variables, though the effects and mitigation strategies differ.

A common criticism of Econometrics is that it sometimes focuses too heavily on the statistical properties of data without sufficiently considering economic theory. This can lead to a model that fits the data well but lacks meaningful interpretation. Moreover, there is a risk that Econometric analysis can identify correlations without adequately addressing causation. Without a clear understanding of causal relationships, the results might be misleading or not useful for policymaking and decision-making.

Now that we have a better understanding of Econometrics and following techniques, let us look at the fundamental ideas of regression modeling in Econometrics.

2. Regression

We know that regression is a statistical technique used to examine the relationship between one dependent variable and one or more independent variables. In statistical terminology, the dependent variable is called endogenous, and the independent is known as exogenous. Linear models are favored in Econometrics modelling. The robustness of linear models contributes to their widespread use in financial analysis and decision-making.

Now, let us discuss about Linear Regression, which is at the center stage of the entire analysis in this book. We shall delve deeper to gain a conceptual understanding and subsequently the mathematical intuition behind regression process.

2.1 Types of regression

A simple Linear Regression involves one dependent variable and one independent variable, which is univariate or bivariate (considering one dependent and one independent) to model the relationship as a straight line. This can be defined through an algebraic formula of a straight line $Y = \beta_0 + \beta_1 X + \epsilon$ where,

- Y = dependent variable,
- β_0 = intercept,
- β_1 = slope,
- X = independent variable and
- ϵ = error term

The above equation can be extended to a multiple linear regression which involves one dependent variable and two or more independent variables as $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$. Here multiple independent variables ($X_1, X_2, \dots, X_n$) and their coefficients ($\beta_1, \beta_2, \dots, \beta_n$) exist.

The fundamental requirement in a Linear Regression is that the relationship between the dependent variable Y and the parameters (β coefficients) is linear. This means that the model is linear in terms of the parameters, even if the variables themselves undergo nonlinear transformations such as X^2 or X^3 etc.

From $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$, the relationship between Y and the parameters $\beta_0, \beta_1, \dots, \beta_n$ is linear. When the variables undergo nonlinear transformations such as polynomials, or exponentials etc., the model can still be linear in terms of the parameters.

Let us look at a few examples:

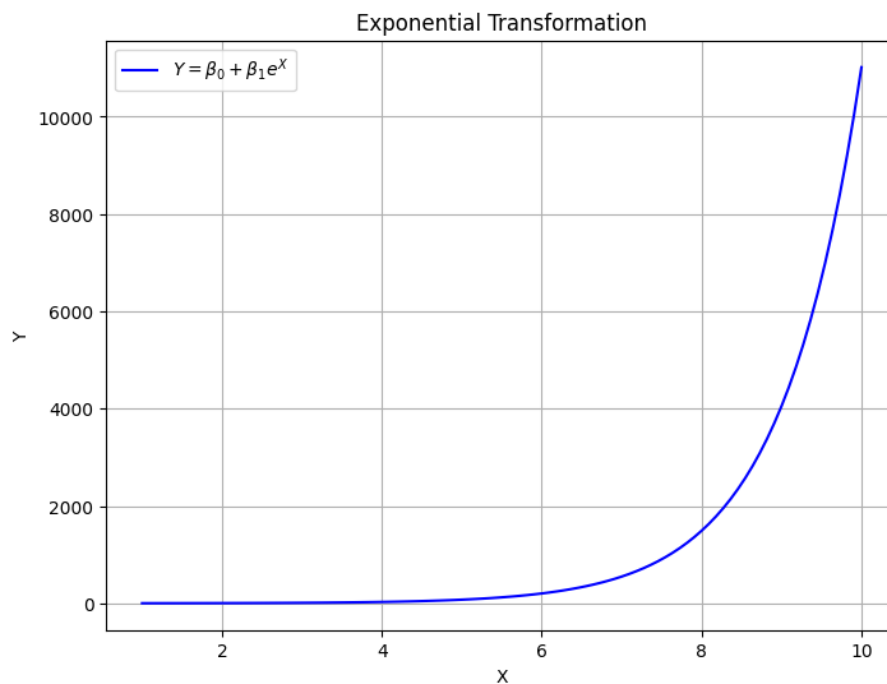
1) Exponential Transformation: Suppose we have a variable Z which is an exponential function of X ($Z = e^X$). The standard linear regression model ($Y = \beta_0 + \beta_1 X + \epsilon$) can be written as $Y = \beta_0 + \beta_1 e^X + \epsilon$, where e^X is a transformed variable, and the relationship with the parameter β_1 remains linear. This means standard Linear Regression techniques can still estimate β_0 and β_1 by minimizing the residuals, even though the relationship between Y and X is non-linear. Since e^X changes non-linearly with X , this results in an exponential response. A one-unit increase in X changes e^X to e^{X+1} . Since $e^{X+1} = e^X * e$, the increase in e^X when X increases by one unit is $\Delta(e^X) = e^{X+1} - e^X = e^X * e - e^X = e^X(e - 1)$

This shows that for a one-unit increase in X , the corresponding change in Y is $\Delta Y = \beta_1 \Delta(e^X) = \beta_1(e^X)(e - 1)$. This means that for each one-unit change in X , the change in Y is proportional to both β_1 and e^X , scaled by $(e - 1)$.

Table 4 displays an exponential curve for the relationship $Y = 1 + 0.5e^X$. We can see here that as X increases, the growth rate of Y increases exponentially. Each unit increase in X results in an increasingly larger increase in Y due to the exponential function.

Table 4. Exponential plot on synthetic data.

```
X = np.linspace(1, 10, 100)
beta_0, beta_1, beta_2 = 1, 0.5, 0.05
Y_exponential = beta_0 + beta_1 * np.exp(X)
plt.figure(figsize = (8, 6))
plt.plot(X, Y_exponential, label = r"$Y = \beta_0 + \beta_1 e^X$", color = "blue")
plt.title("Exponential Transformation")
plt.xlabel("X")
plt.ylabel("Y")
plt.legend()
plt.grid(True)
plt.show()
```



This non-linear effect of X means, while the model is linear in its parameters (making it feasible for the Linear Regression methods), the interpretation of β_1 adapts to reflect the exponential transformation applied to X .

2) Logarithmic Transformation: In the case of logarithmic transformation, the model is $Y = \beta_0 + \beta_1 \log(X) + \epsilon$.

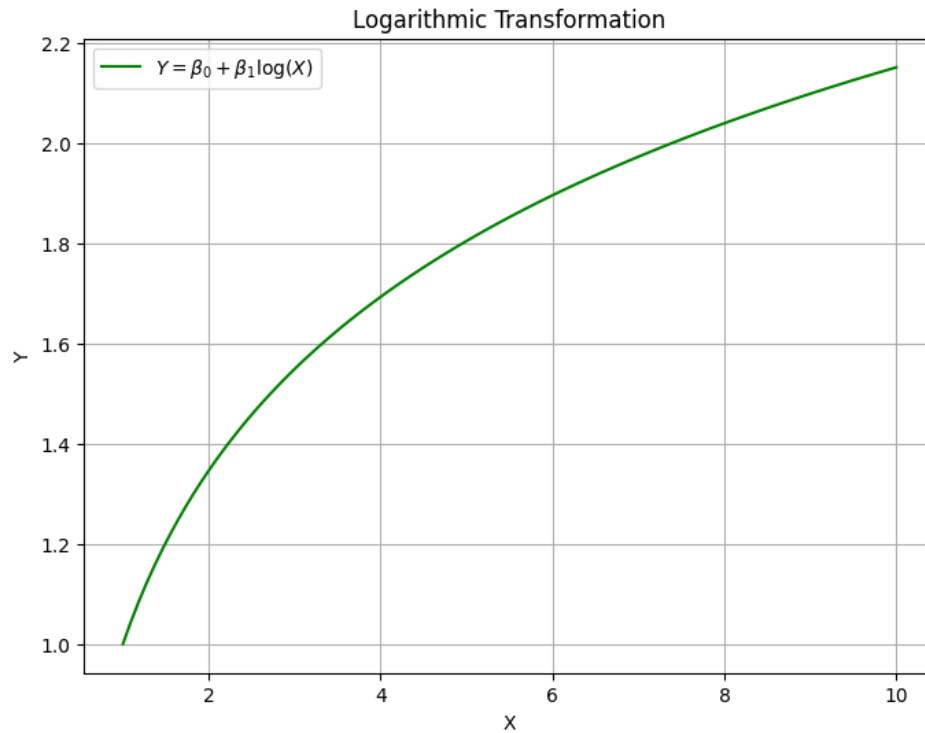
The interpretation of β_1 changes here, β_1 now is the expected change in Y for a one-percent change in X , assuming other factors remain constant. Specifically, β_1 is the change in Y for a one-unit change in $\log(X)$, rather than a one-unit change in X itself. A one-unit increase in $\log(X)$ corresponds approximately to a percentage change in X when X changes proportionally.

β_1 can be interpreted as the elasticity of Y with respect to X . In economics, this elasticity measures the percentage change in one variable (in this case, Y) in response to a 1% change in another variable (here, X). When β_1 is the coefficient of $\log(X)$, it suggests that a 1% increase in X results in approximately a $\beta_1\%$ change in Y .

Table 5 displays a logarithmic transformation and the relationship between X and Y using a model of the form $Y = \beta_0 + \beta_1 \log(X)$. The plot shows a logarithmic growth curve. As X increases, the increase in Y becomes smaller, meaning Y grows more. Moreover, each unit increase in X results in a smaller incremental change in Y compared to an exponential or linear function.

Table 5. Logarithmic transformation and visualization.

```
X = np.linspace(1, 10, 100)
beta_0, beta_1, beta_2 = 1, 0.5, 0.05
Y_logarithmic = beta_0 + beta_1 * np.log(X)
plt.figure(figsize=(8, 6))
plt.plot(X, Y_logarithmic, label=r"$Y = \beta_0 + \beta_1 \log(X)$", color="green")
plt.title("Logarithmic Transformation")
plt.xlabel("X")
plt.ylabel("Y")
plt.legend()
plt.grid(True)
plt.show()
```



This interpretation is useful in the economic and business contexts where understanding proportional relationships, such as sensitivity or responsiveness, is valuable. A one-unit change in $\log(X)$ approximates a proportional change in X when X increases multiplicatively. However, the logarithmic transformation is still considered linear because the model retains the linearity in its parameters rather than the variables.

3) Polynomial Transformation: In the case of a polynomial transformation, the model

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$$

This model introduces both a linear and a quadratic component of X , enabling the model to capture curvature in the relationship between X and Y . β_1 represents the linear effect of X on Y . It shows the change in Y for a one-unit change in X , holding the X^2 term constant. This linear term provides the base or initial slope of the relationship between X and Y and is like the slope in a simple Linear Regression.

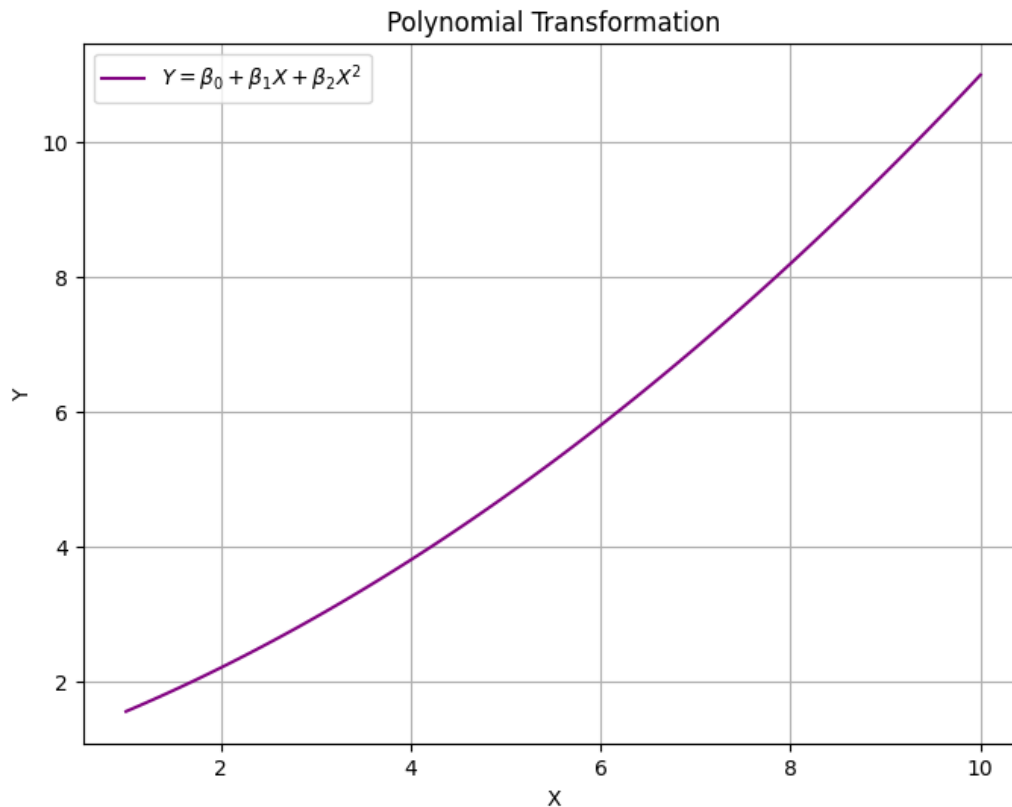
β_2 represents the quadratic effect of X on Y . It captures the rate at which the slope changes as X changes, introducing curvature into the model. If β_2 is positive, Y increases at an increasing rate as X increases, resulting in an upward curve. Conversely, if β_2 is negative, Y initially increases but at a decreasing rate, creating a downward curve as X continues to grow.

The combination of linear and quadratic terms allows the model to better fit non-linear relationships. In contexts where Y does not change at a constant rate with X , this model can represent the data better than a purely linear model. Together, β_1 and β_2 capture both the direction and shape of the relationship between X and Y . This flexibility is helpful in modeling complex relationships where Y may increase or decrease in a curved fashion as X changes.

Table 6 displays a polynomial transformation, where Y is modeled as a quadratic function of X , using the equation $Y = \beta_0 + \beta_1 X + \beta_2 X^2$. The plot shows a parabolic curve, illustrating a polynomial (quadratic) transformation. The inclusion of X^2 introduces a curved relationship, creating a parabolic shape that either opens upward or downward based on β_2 . As X grows, the increase in Y becomes larger due to the X^2 term, resulting in an accelerating effect on Y for positive values of β_2 .

Table 6. Polynomial transformation and visualization.

```
X = np.linspace(1, 10, 100)
beta_0, beta_1, beta_2 = 1, 0.5, 0.05
Y_polynomial = beta_0 + beta_1 * X + beta_2 * X**2
plt.figure(figsize = (8, 6))
plt.plot(X, Y_polynomial, label=r"$Y = \beta_0 + \beta_1 X + \beta_2 X^2$", color = "purple")
plt.title("Polynomial Transformation")
plt.xlabel("X")
plt.ylabel("Y")
plt.legend()
plt.grid(True)
plt.show()
```

The inclusion of X^2 does not change the linearity of the model in the parameters because it is simply an additional variable in the regression equation. Therefore, the polynomial model remains a type of linear regression.

Let us tabulate the importance of regression as displayed in **Table 7** for better understanding and memorize.

Table 7. Importance of regression in Econometrics.

Important factors	Description
Quantifying relationships	Regression analysis helps quantify the strength and direction of relationships between economic variables. For example, understanding how changes in interest rates affect investment levels.
Testing economic theories	Econometricians use regression analysis to test economic theories and hypotheses.
Forecasting	Regression models can be used to make predictions about future values of variables based on historical data. For example, predicting future GDP growth based on factors such as investment, consumption, and government spending.
Policy evaluation	Policymakers use regression analysis to evaluate the potential impact of different policy measures. For example, assessing the impact of a tax cut on consumer spending.

Identifying causal relationships	A correlation does not mean that the change in one variable is the cause of the change in the values of the other variable. There is a causal relationship between the two events, meaning that one event is the consequence of the other event occurring. The statistical concept of causation is often misinterpreted and applied incorrectly, leading one to assume that just because data exhibits a correlation, there must be an underlying causative relationship.
Handling complexity	Regression analysis allows economists to handle complex relationships involving multiple variables simultaneously. This is crucial for understanding real-world economic phenomena where multiple factors interact.

2.1.1 Ordinary Least Square (OLS) regression

Let us demonstrate how to estimate the parameters of linear regression using Ordinary Least Squares (OLS) regression works.

From $Y = \beta_0 + \beta_1 X_i + \varepsilon$, the values of Y and X can be easily computed from the data set. However, that is not the case with β_0 , β_1 and ε . Here we use OLS to make an estimate based on what their values are. We will use $\hat{\beta}_0$ and $\hat{\beta}_1$ to denote the model estimates. OLS estimates the parameters $(\beta_0, \beta_1, \dots, \beta_n)$ by minimizing the Residual Sum of Squares (RSS).

$$RSS = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Here, Y_i is actual or observed value and $\hat{Y}_i$ is the predicted value. By minimizing RSS, OLS finds the best-fitting line that explains the relationship between the independent and dependent variables.

We already know now that in quantitative finance, the data sets are small. Moreover, data are mostly numerical and have a low signal-to-noise ratio. Such as stock prices can be influenced by many factors, including market sentiment, macroeconomic factors, and volatility which makes it challenging to extract meaningful signals. Due to these, the statistical estimates (e.g., estimate of returns, volatility, correlations) often come with high uncertainty. This can lead to a higher margin of errors in estimates.

Before applying OLS regression, it is crucial to ensure the assumption of linearity between the independent variable (X) and the dependent variable (Y). We must check if the relationship between X and Y is indeed linear. **Table 8** displays the spot check on linearity.

Table 8. Data generation and Spot check on linearity.

```
np.random.seed(0)
X = np.random.rand(100, 1) * 10
Y = 2.5 * X + np.random.randn(100, 1) * 2
data = pd.DataFrame({'X': X.flatten(), 'Y': Y.flatten()})
plt.scatter(data['X'], data['Y'])
plt.title('Scatter Plot of X vs Y')
plt.xlabel('X')
plt.ylabel('Y')
plt.grid()
plt.show()
```

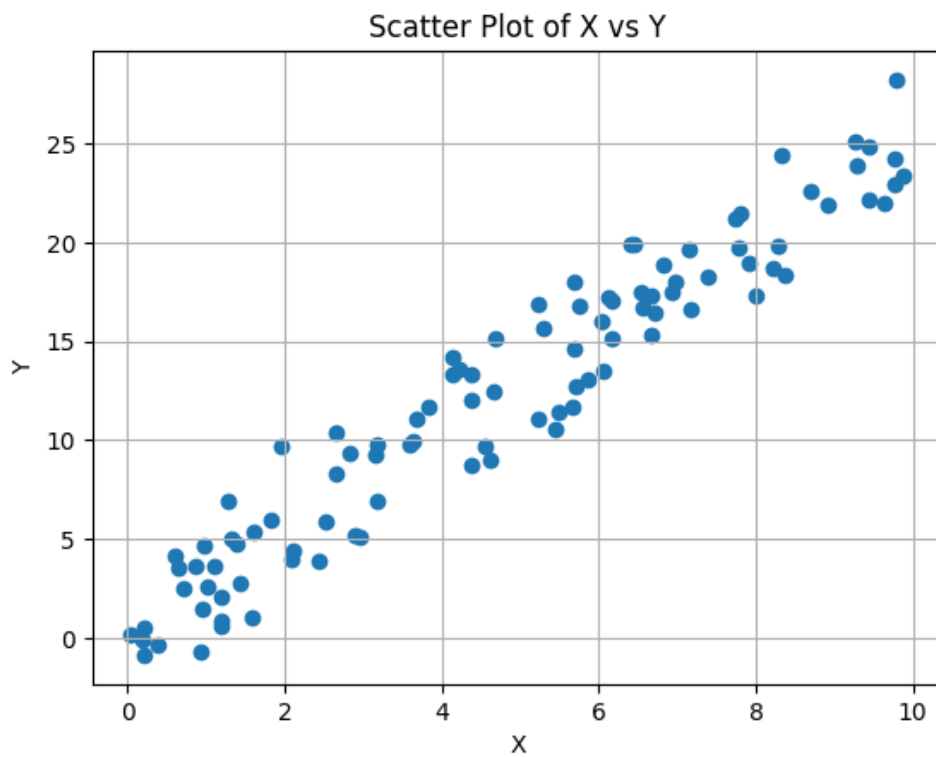


Table 9. OLS model fit.

```
X_with_const = sm.add_constant(X)
model = sm.OLS(Y, X_with_const).fit()
print(model.summary())
```

```

=====
                        OLS Regression Results
=====
Dep. Variable:          y      R-squared:          0.928
Model:                  OLS    Adj. R-squared:       0.928
Method:                 Least Squares    F-statistic:      1270.
Date:                   Fri, 12 Jul 2024    Prob (F-statistic): 6.76e-58
Time:                   05:52:21    Log-Likelihood:   -210.83
No. Observations:      100    AIC:              425.7
Df Residuals:          98    BIC:              430.9
Df Model:               1
Covariance Type:        nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	0.4443	0.387	1.149	0.253	-0.323	1.211
x1	2.4874	0.070	35.630	0.000	2.349	2.626

```

=====
Omnibus:                 11.746    Durbin-Watson:       2.083
Prob(Omnibus):           0.003    Jarque-Bera (JB):     4.097
Skew:                    0.138    Prob(JB):             0.129
Kurtosis:                 2.047    Cond. No.             10.9
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

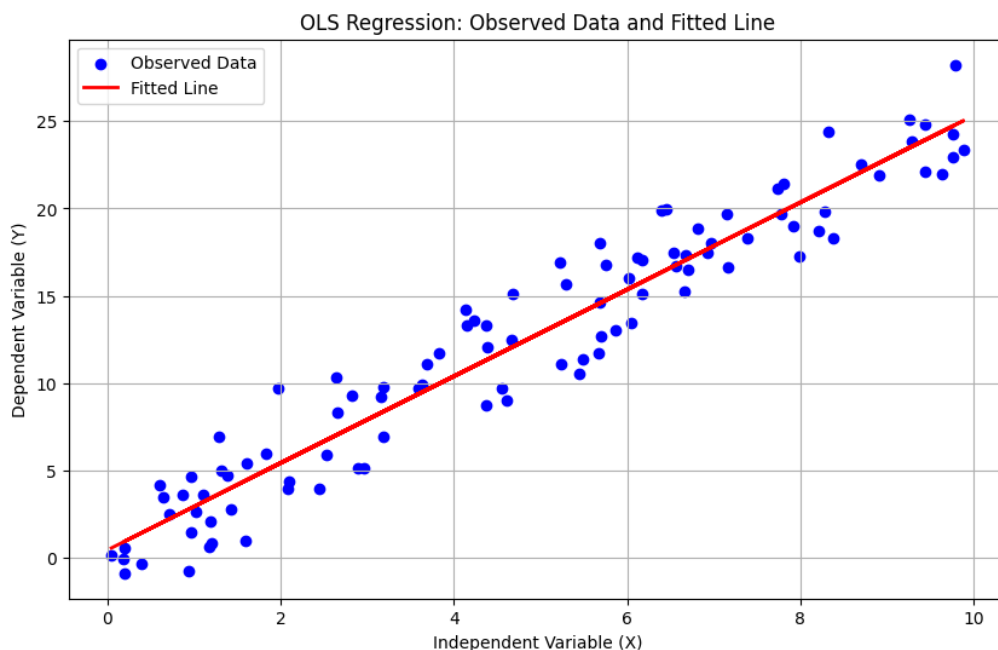
We get a bunch of output when we run OLS. Let us understand as how to interpret the generated output.

- I. R^2 (0.928) indicates that approximately 92.8% of the variation in the dependent variable Y can be explained by the independent variable X. This suggests a good fit of the model.
- II. The intercept term (const) is 0.4443, which means that when the independent variable X is zero, the expected value of the dependent variable Y is 0.4443. However, the p-value for this coefficient (0.253) > 0.05 means it is statistically not significant.
- III. The slope coefficient (x1) is 2.4874, meaning that for every one unit increase in X, Y increases by approximately 2.4874 units. This coefficient is highly significant (p-value = 0.000).
- IV. F-statistic (1270) and Prob (F-statistic): The high F-statistic and its associated low p-value (6.76e-58) indicate that the model is statistically significant, and that the independent variable X has a strong linear relationship with the dependent variable Y.
- V. The Omnibus Test and Jarque-Bera Test suggest that there may be some deviation from normality in the residuals, though not extreme. The Durbin-Watson statistic (2.083) is close to 2, indicating that there is no significant autocorrelation in the residuals.

This is a simple example of the implementation and interpretation of the OLS model. We shall apply this knowledge in the subsequent sections where we explore more into the OLS regression model. **Table 10** displays how to generate a line fit plot from the above model.

Table 10. Line fit plot.

```
data['Predicted_Y'] = model.predict(X_with_const)
plt.figure(figsize = (10, 6))
plt.scatter(data['X'], data['Y'], color = 'blue', label = 'Observed Data')
plt.plot(data['X'], data['Predicted_Y'], color = 'red', label = 'Fitted Line', linewidth = 2)
plt.title('OLS Regression: Observed Data and Fitted Line')
plt.xlabel('Independent Variable (X)')
plt.ylabel('Dependent Variable (Y)')
plt.legend()
plt.grid(True)
plt.show()
```



OLS assumes linearity, which means that the relationship between the independent variable X and the dependent variable Y can be represented by a straight line. Above plot displays the line fit plot from the OLS regression analysis. The blue dots are the data points, and the red line is the fitted line from the model. This plot visually demonstrates the relationship between the independent variable X and the dependent variable Y , along with the best-fit line being the sum of squared residuals. The process involves squaring the vertical distance between

each point and the regression line and then minimizing the total sum of these squared distances (hence "least squares").

Let us now summarize and tabulate our understanding on OLS as shown below.

Reason	Description
Minimize errors	By minimizing the sum of the squared residuals (SSR), we ensure that the overall error between the observed and predicted values is minimized.
Positive and negative deviations	Squaring the residuals ensures that both positive and negative deviations from the regression line are treated equally. Without squaring, positive, and negative residuals could cancel each other out, leading to a misleading representation of the error.
Mathematical simplicity	The least squares method results in a quadratic optimization problem, which is easier to solve mathematically. The resulting normal equations, derived from setting the derivative of the sum of squared residuals to zero, provide a straightforward way to find the regression coefficients.
Statistical properties	Under the assumptions of the classical linear regression model (e.g., linearity, independence, homoscedasticity, normality of errors), the least squares estimators have desirable statistical properties. They are unbiased, consistent, and efficient, meaning they have the smallest variance among all linear unbiased estimators (known as the Gauss-Markov theorem).

2.1.2 Mathematical explanation

We are familiar with the simple linear regression formulation. Considering the formula from $Y = \beta_0 + \beta_1 X_i + \varepsilon$.

- The residual for each observation is: $e = Y - (\beta_0 + \beta_1 X_i)$.
- Our goal is to minimize the RSS: $\sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n [Y_i - (\beta_0 + \beta_1 X_i)]^2$. To find the best β_0 and β_1 , we solve the equations derived from setting the derivatives of the SSR to zero $\left(\frac{\partial SSR}{\partial \beta_0} = 0, \frac{\partial SSR}{\partial \beta_1} = 0 \right)$.

It is important to note that while the relationship being shown as a straight line, the model needs to be linear in the parameters (e.g., β_0 and β_1). This means that the parameters β_0 (intercept) and β_1 (slope) should not be multiplied together, divided, squared, or raised to any other power. We have discussed that considering the parameters (coefficients) β_0 and β_1 are in a straight-line form, such as, $\beta_0 + \beta_1 X$, the variables Y and X can still be transformed in other

ways (e.g., logarithms, polynomials) as long as the relationship remains linear in the parameters such as $Y = \beta_0 + \beta_1 \log(X)$ or $Y = \beta_0 + \beta_1 X^2$.

There are various terms and concepts we encounter when learning about regression and Econometrics. In **Table 11** we present the relevant terminologies when explaining about Econometrics and regressions.

Table 11. Regression terminologies.

Terminology	Mathematical expression	Explanation
Dependent variable	Y	The outcome variable that the model aims to predict.
Independent variable	X	The predictor variable used to explain variations in the dependent variable.
Coefficient	β	Is the change in the dependent variable for a one-unit change in the independent variable.
Intercept	β_0	The expected value of the dependent variable when all independent variables are zero.
Error term	ε	Difference between the observed and predicted values of the dependent variable, $\varepsilon_i \sim N(0, \sigma^2)$ are i.i.d (independent and identically distributed).
Simple linear regression or Ordinary least Square (OLS) regression.	$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$, where $i = 1, 2, \dots, n$ for each of the n observations.	Regression model with one dependent variable and one independent variable.
Multivariate regression.	$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_n X_{ni} + \varepsilon_i$	Regression model with one dependent variable and multiple independent variables.
R-squared	R^2	The proportion of variance in the dependent variable explained by the independent variables.
Adjusted R squared	$Adj R^2$	Adjusted for the number of predictors in the model, providing a better measure of fit.
Ordinary least square	$\min \sum (Y_i - \hat{y}_i)^2$	A method for estimating the coefficients by minimizing the sum of the squared residuals.
F-statistics	$F = \frac{\text{explained variance}}{\text{unexplained variance}}$	Tests the overall significance of the regression model.
Standard error	SE	The standard deviation of the sampling distribution of a statistic, typically a coefficient.
t-statistics	$t = \frac{\beta}{SE(\beta)}$	Tests the significance of individual coefficients.

p-value		The probability of observing the test results under the null hypothesis.
Confidence interval	$\hat{\beta} \pm t^* * SE(\hat{\beta})$	The range within which the true parameter value is expected to fall with a certain probability.
Multicollinearity		Where independent variables are highly correlated, potentially distorting coefficient estimates.
Homoscedasticity		The assumption that the variance of the error terms is constant across observations.
Heteroscedasticity		The condition where the variance of the error terms varies across observations.
Auto correlation	ρ	The correlation of a variable with itself over successive time intervals.
Durbin-Watson Statistic	$DW = \frac{\sum (e_t - e_{t-1})^2}{\sum e_t^2}$	Used to detect the presence of autocorrelation in the residuals.
Dummy Variable		A binary variable used to include categorical data in regression models.
Instrumental Variable	Z	A variable used to account for endogeneity, not correlated with the error term.
Lagged variable	X_{t-1}	It is the prior time period's value used in time series analysis.
Cointegration		Statistical property of time series variables that move together in the long run.
Stationary		Property of a time series where mean and variance are constant over time.

3. Timeseries data

This section addresses the fundamental aspects of handling the data. Here, we delve deeper into the timeseries data, which is the emphasis of this book. We will learn how to ingest data and perform some basic yet significant transformations that will come in handy during real-world analysis. Let us familiarize ourselves with time series data.

Table 12 shows the data ingestion from Yahoo Finance⁴ and line plots showing Tesla's daily stock price from 2018 till 2022 and, subsequently **Table 13** compares each year's prices.

Table 12. Time series data.

```
data = yf.download('TSLA', start = '2018-01-01', end = '2023-01-01')['Close']
data.plot(grid = True)
```

⁴ <https://finance.yahoo.com/quote/TSLA/history/>


```
plt.show()
```

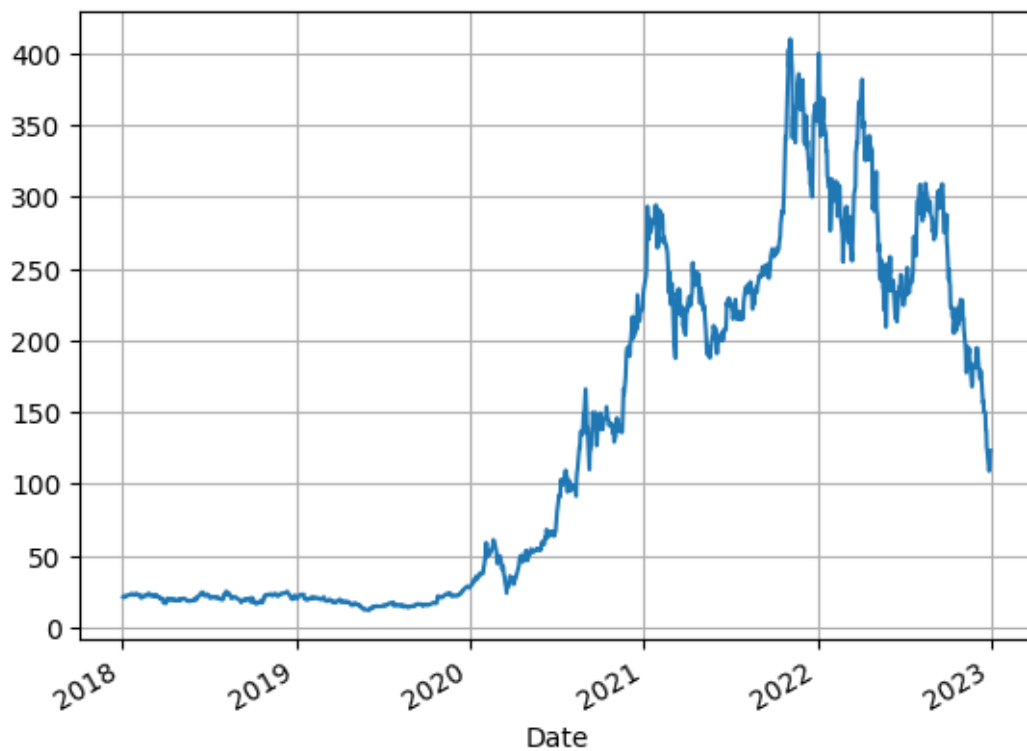
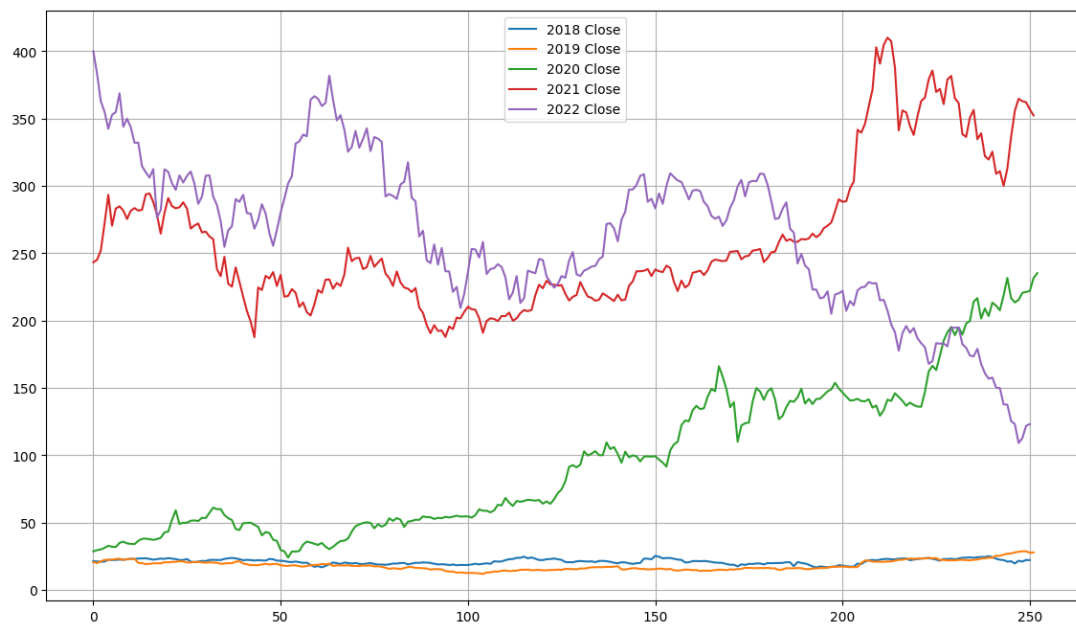


Table 13. Compare each years Tesla stock price.

```
tesla = pd.DataFrame()
for year in ["2018", "2019", "2020", "2021", "2022"]:
    price_per_year = data.loc[year].reset_index(drop=True)
    price_per_year.name = year + " Close"
    tesla = pd.concat([tesla, price_per_year], axis=1)
print(tesla.head())
```

	2018 Close	2019 Close	2020 Close	2021 Close	2022 Close
0	21.368668	20.674667	28.684000	243.256668	399.926666
1	21.150000	20.024000	29.534000	245.036667	383.196655
2	20.974667	21.179333	30.102667	251.993332	362.706665
3	21.105333	22.330667	31.270666	272.013336	354.899994
4	22.427334	22.356667	32.809334	293.339996	342.320007

```
tesla.plot(figsize = (14, 8), grid = True)
plt.show()
```



We can convert tesla stock price to monthly frequency by using the following code and **Figure 5** displays the output.

```
data = yf.download('TSLA', start='2018-01-01', end='2023-01-01')['Close']
data.asfreq("M").plot(grid=True)
plt.show()
```

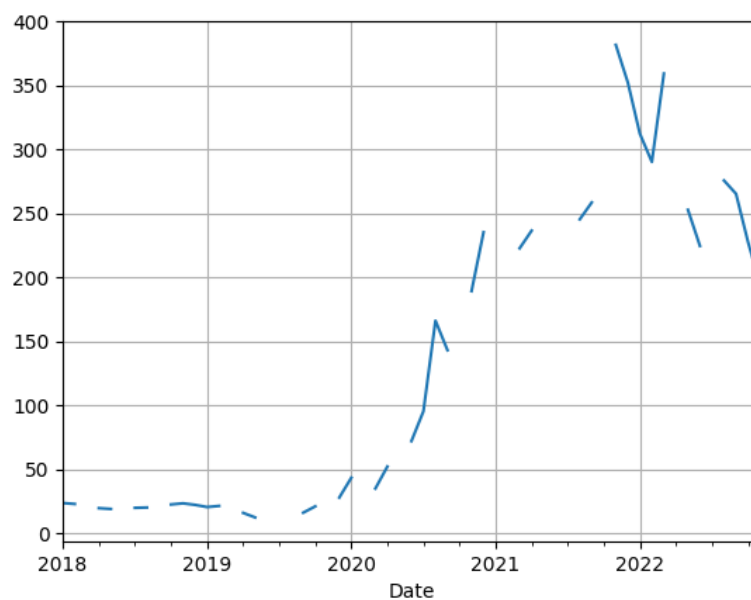


Figure 5. Tesla stock price in monthly frequency.

3.1 Lagged values & daily changes

Lagged values in timeseries help in understanding how current values depend on the past values. This is crucial in finding patterns and trends within the data and subsequently developing a dynamic model. We will discuss the implementation of dynamic modeling in Error! Reference source not found..

Moreover, lagged values can be used in the stationarity tests like the Augmented Dickey-Fuller (ADF) test. Stationarity is a fundamental assumption in many time series models. ADF tests the null hypothesis showing the presence of unit root in a time series. The alternative hypothesis is that the series is stationary.

In price series, returns are estimated using daily price differences, which also helps estimate volatility. High volatility often indicates greater risk and uncertainty. **Table 14** displays the necessary transformation being applied to the Tesla stock price.

Table 14. Lagged values and daily changes.

```
data = yf.download('TSLA', start = '2018-01-01', end = '2023-01-01')['Close']
tesla = pd.DataFrame(data)
tesla['Lag_1'] = tesla['Close'].shift(1)
tesla['Daily_Change'] = tesla['Close'] / tesla['Lag_1']
tesla.head()
```

	Close	Lag_1	Daily_Change
Date			
2018-01-02	21.368668	NaN	NaN
2018-01-03	21.150000	21.368668	0.989767
2018-01-04	20.974667	21.150000	0.991710
2018-01-05	21.105333	20.974667	1.006230
2018-01-08	22.427334	21.105333	1.062638

3.2 Logarithmic transformation

Logarithmic transformation is widely used in economics and finance because it aligns closely with how these fields conceptualize relationships between variables, stabilizes variance,

reduces skewness, and provides intuitive interpretations in terms of percentages and elasticity. Many economic and financial relationships are nonlinear in nature. A log transformation can linearize these relationships, making them easier to model using traditional linear regression techniques. Moreover, Heteroscedasticity (non-constant variance of errors) is common in economic and financial data, where variability increases with the size of the variable. Log transformation can stabilize the variance, improving model reliability and the validity of statistical inferences. In financial data, variables like income, market capitalization, or stock prices often span several orders of magnitude. A log transformation compresses large values and expands small ones, making the data easier to visualize, interpret, and model. It also helps reducing the skewness in the data.

Let us examine the advantages of logarithmic transformation and its use. A time series normally shows traces of growth or decline, i.e., non-stationarity. When the characteristics of a time series, such as mean, variance, or covariance, alter with time, it is said to be non-stationary. This shows that prior observations might not be indicative of the future and that the data's behavior is neither steady nor predictable. Non-stationary time series cannot be used in regression models because they may create spurious regression, which is a false relationship.

A non-constant variance series can be stabilized using log transformation. We must remember that log transformation can only be used with positively valued time series. Taking a log reduces the values towards zero. Values close to 1 shrink less, while values higher shrink more, minimizing variance.

Suppose Y_t is an observation of the time series in period t , the growth rate (G_t) from period $t - 1$ is $G_t = \frac{Y_t}{Y_{t-1}}$. If we rearrange this equation and take natural log we derive at
$$l_n(1 + G_t) = l_n\left(\frac{Y_t}{Y_{t-1}}\right) = l_n Y_t - l_n Y_{t-1}.$$

If we want to understand what is, $l_n(1 + G_t)$, we must go back to our fundamental calculus and Taylor's expansion to approximate any functions as
$$l_n(1 + X) = X - \frac{1}{2}X^2 + \frac{1}{3}X^3 - \frac{1}{4}X^4 + \dots = \sum_{k=1}^{\infty} (-1)^{k+1} \frac{X^k}{k}.$$

Because the growth rate is generally small, we can use the linear term exclusively in Taylor expansion ($l_n(1 + X) \approx X$) which means log difference approximates the growth rate ($l_n Y_t - l_n Y_{t-1} \approx G_t$).

Table 15 shows how to prepare GDP data for detailed time series analysis, including the creation of lagged variables, calculating growth rates, and applying log transformations. These steps are essential for understanding trends, making forecasts, and performing various statistical analyses on economic data.

Table 15. Transformation of time series data.

```
gdp = pd.read_csv('/content/GDP.csv')

gdp['DATE'] = pd.to_datetime(gdp['DATE'])
gdp.set_index('DATE', inplace = True)
gdp['Lag_1'] = gdp['GDP'].shift(1)
gdp['log_gdp'] = np.log(gdp['GDP'])
gdp["growth_rate"] = gdp["GDP"] / gdp["Lag_1"]
gdp["growth_rate"] = gdp["GDP"] - 1
gdp["log_approx"] = np.log(gdp["GDP"]) - np.log(gdp["Lag_1"])

gdp.dropna(inplace = True)
gdp.head(10)
```

	GDP	Lag_1	log_gdp	rate	log_approx
DATE					
2010-04-01	14980.193	14764.610	9.614484	0.014601	0.014496
2010-07-01	15141.607	14980.193	9.625202	0.010775	0.010718
2010-10-01	15309.474	15141.607	9.636227	0.011086	0.011025
2011-01-01	15351.448	15309.474	9.638965	0.002742	0.002738
2011-04-01	15557.539	15351.448	9.652301	0.013425	0.013336
2011-07-01	15647.680	15557.539	9.658078	0.005794	0.005777
2011-10-01	15842.259	15647.680	9.670436	0.012435	0.012358
2012-01-01	16068.805	15842.259	9.684635	0.014300	0.014199
2012-04-01	16207.115	16068.805	9.693206	0.008607	0.008571
2012-07-01	16319.541	16207.115	9.700119	0.006937	0.006913

The log transformation is an effective approximation technique for small growth rates ($< 10\%$). This is due to the property of logarithms where $\log(1 + x) \approx x$ when x is small. However, as growth rates increase, this approximation becomes less correct. For larger growth rates, the log transformation introduces a noticeable error because the approximation $\log(1 + x) \approx x$ becomes less precise. In such situations, alternative methods, such as directly modeling exponential functions or using more sophisticated techniques, are needed to capture the growth rate accurately. This understanding is crucial when applying the natural log transformation in financial modeling and time series analysis to ensure the accuracy of the results. **Table 16** displays the reliability issue with log transformation.

Table 16. Reliability of natural logarithm.

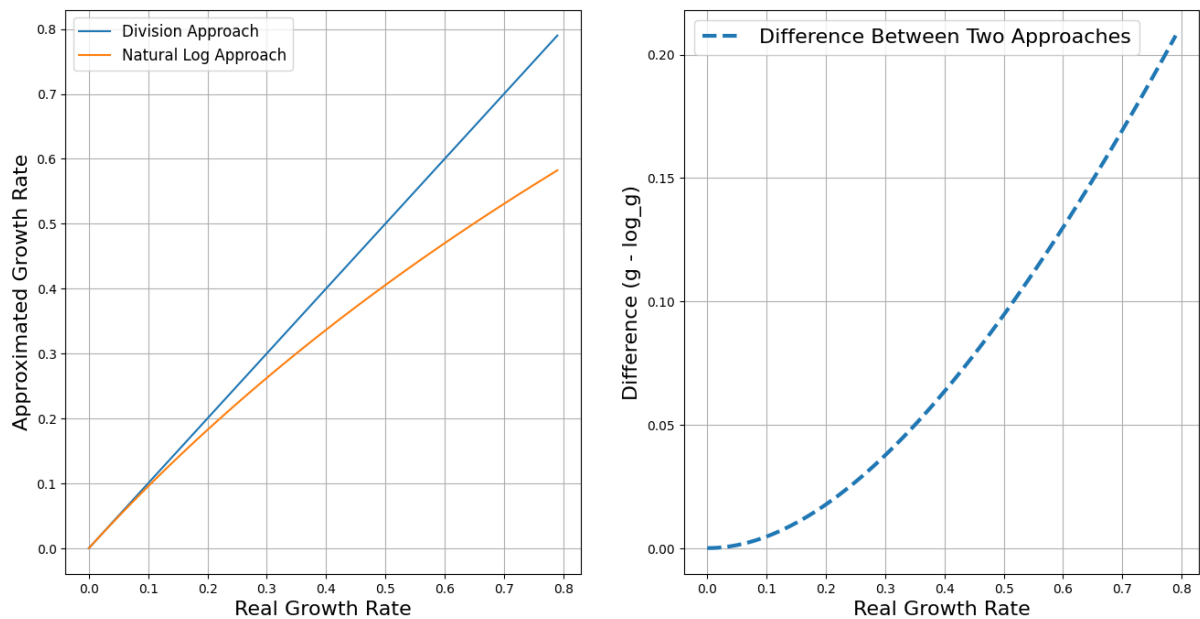
```
g = np.arange(0, 0.8, 0.01)
log(g) = np.log(1 + g)
fig, ax = plt.subplots(nrows = 1, ncols = 2, figsize = (16, 8))
ax[0].plot(g, g, label = "Division Approach")
ax[0].plot(g, log_g, label = "Natural Log Approach")
ax[0].set_xlabel("Real Growth Rate")
ax[0].set_ylabel("Approximated Growth Rate")
ax[0].grid()
```

```

ax[0].legend()

ax[1].plot(g, g - log_g, ls="--", lw=3, label="Difference Between Two Approaches")
ax[1].set_xlabel("Real Growth Rate")
ax[1].set_ylabel("Difference (g - log_g)")
ax[1].grid()
ax[1].legend()
plt.show()

```



The first plot on the left-hand side shows the real growth rate versus the approximated growth rate using both the division approach and the natural log approach. For small growth rates (close to 0), the lines overlap, showing that the natural log approach is a good approximation. However, as the growth rate increases, the discrepancy between the two approaches becomes more apparent. The second plot on the right-hand side shows the difference between the actual growth rate and the log-transformed growth rate. It shows that for growth rates less than approximately 0.1 (10%), the error introduced by using the log approach is minimal and acceptable. Beyond this point, the error grows, and the log transformation becomes less reliable.

4. Key takeaways

To this end, this chapter introduces fundamental concepts of Econometrics. Econometrics integrates economic theory, mathematical modeling, and statistical inference to analyze real-world economic data. It provides a structured approach to understanding complex economic

phenomena, such as consumer behavior, economic growth, inflation, and market dynamics. Econometrics helps in making predictions and policy decisions by quantifying relationships between variables. The history of Econometrics began with Frisch in 1926, with Tinbergen and Klein and Kuznets all contributing significantly to the field. The first step in Econometrics is to obtain and analyze a dataset and define hypotheses based on existing economic theories. Modeling helps clarify the structure of relationships and test hypotheses.

Models provide a structured method for testing hypotheses, quantifying relations, forecasting future trends, and making evidence-based decisions in economic policy and business strategy. By using a model, we evaluate economic ideas and challenge assumptions, allowing for quantification of relationships between variables. Moreover, models are essential for forecasting, strategic planning, and policy evaluation. This chapter discusses model testing using ordinary least squares regression technique.

References

- [1] Klein, L. R. (1969). The specification of regional Econometric models. *Papers of the Regional Science Association*, 23(1), 105–115. <https://doi.org/10.1007/BF01941877>
- [2] Simon Kuznets. (1971). *Modern Economic Growth: Findings and Reflections* . Simon Kuznets Prize Lecture.